

9.4: Divergence and Integral Tests

Calculus III | MTH 331 | Section A | Fall 2012

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- Also, if $\sum_{k=0}^{\infty} a_k$ converges/diverges then $\sum_{k=m}^{\infty} a_k$ also converges/diverges.

The Divergence Test

Theorem: Divergence Test

- a. If $\sum a_k$ converges, then $\{a_k\}^\infty = 0$ ($\lim_{k \rightarrow \infty} a_k = 0$).
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- The Divergence Test allows us to evaluate a series by looking at a sequence.

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If f is a continuous, positive, decreasing function for $x \geq 1$ such that $a_k = f(k)$ for $k = 1, 2, \dots$ then $\sum_{k=1}^{\infty} a_k$ and $\int_1^{\infty} f(x)dx$ either **both** converge or diverge.

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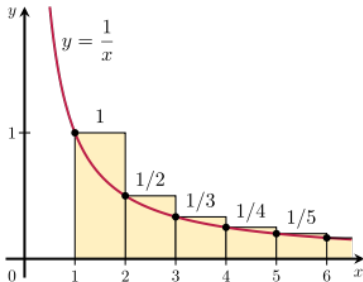
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$$\int_2^{\infty} \frac{1}{x \ln x} dx = \lim_{b \rightarrow \infty} \ln(\ln x) \Big|_2^b = \lim_{b \rightarrow \infty} [\ln(\ln b) - \ln(\ln 2)] = \infty$$

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While in general when $\int a_k$ and $\sum a_k$ converge it is to different values, we can *estimate* $\sum a_k$ using $\int a_k$.

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Theorem: Estimating Series with Positive Terms

Let f be positive, decreasing, and continuous for $x \geq 1$ and $f(k) = a_k$ for $k = 1, 2, \dots$. If $\sum_{k=1}^{\infty} a_k$ converges to S then,

$$S_n + \int_{n+1}^{\infty} f(x) dx \leq \sum_{k=1}^{\infty} a_k \leq S_n + \int_n^{\infty} f(x) dx.$$

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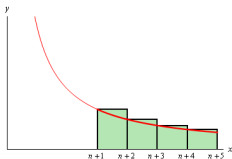
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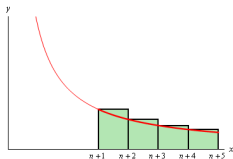
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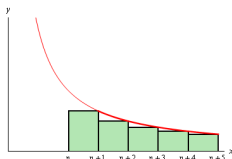
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$$R_n \leq \int_n^{\infty} f(x) dx$$

ex. Using $n = 10$, estimate the value of $\sum_{n=1}^{\infty} \frac{1}{n^2}$
(it is known to be $\frac{\pi^2}{6} \approx 1.64493$.)

